G P S SAI KRISHNA

AI / ML | Computer Vision | Full-Stack | UI / UX Design

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PROFESSIONAL SUMMARY

Computer Vision engineer (B.Tech CSE 2027, CGPA 8.5) who owns the whole ML lifecycle: **dataset curation, model training, evaluation, API serving and app delivery**. Trained a YOLOv8 weapon detector to **~82% mAP@50** on 137K images and deployed it on a CCTV and autonomous-drone network, earning **1st prize at three consecutive project expos, including INSPIRA '26 at IIT Hyderabad**. Also ships production full-stack software for paying clients (React, Node.js, Firebase).

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript (ES6+), Dart, Java, C, SQL
AI / Computer Vision	PyTorch, Ultralytics YOLOv8 / YOLO11, DeepLabV3, OpenCV, MediaPipe, scikit-learn; object detection & segmentation, model evaluation (mAP, Dice), dataset curation & annotation
Speech / LLM	faster-whisper, openWakeWord, speaker verification, LLM API integration
Backend & Data	FastAPI, Node.js, Express.js, REST APIs, JWT auth, PostgreSQL, Supabase, Firebase / Firestore, MongoDB, SQLite
Frontend & UI/UX	React, Next.js, Vue.js, Vite, Tailwind CSS, Three.js, Framer Motion, Figma; Flutter (Android + Windows); Tauri
Embedded / IoT	ESP32 / ESP32-S3 (ESP-IDF, C), BLE, CoAP, MQTT, Crazyflie CRTP / cflib
Cloud & Tools	Git / GitHub, Vercel, Cloudflare, Docker, pytest, Kaggle GPU training · DSA, OOP, DBMS, OS, Computer Networks

EXPERIENCE

Freelance Full-Stack Developer · ARC LABS (client)
Jun 2026 – Present

React · Express.js · Firebase · Razorpay · Vercel · Three.js · Tauri

- Delivered **arclabs.in end to end** for a Hyderabad STEM-lab company: an ~18-page site plus an **admin CRM dashboard** on React, Express and Firebase (Auth, Firestore, Cloud Functions), deployed with Git-based CD on Vercel.
- Integrated **Razorpay payments**, email notifications and QR / PDF generation for a self-serve booking and checkout flow.
- Implemented a full **SEO and social-sharing layer** (canonical URLs, Open Graph, Twitter Cards) and a mobile-first 3D UI.
- Built the **internal ops portal** (22 pages: payroll, P&L, sales CRM, invoicing) as an auto-updating Tauri desktop app.

Full Stack Development Intern · IVIS Labs
Jan 2023 – Apr 2023

- Built an **e-commerce web app from scratch** in a four-month MEVN internship (Vue.js, Express.js, MongoDB, Vite).

PROJECTS

Shadow Falcons: AI Surveillance Drone & CCTV Security Network
1ST PRIZE ×3 — Feb 2026 – Present

Python · PyTorch · Ultralytics YOLOv8 · OpenCV · ESP32-S3 · Crazyflie cflib

- Trained a custom **YOLOv8s detector** on a **137,000-image** firearm and knife dataset: **~82% mAP@50**, ~82% precision, ~74% recall.
- Engineered a **real-time inference pipeline**: three concurrent YOLO models on a Wi-Fi MJPEG stream, run off-board so full-size models work against a palm-sized drone.
- Cut false alarms with a **three-stage filter** (screen-overlap rejection, weapon-to-person association, 3-frame confirmation); live HUD grades threats HIGH / MEDIUM / LOW.
- Flew an ESP32-S3 airframe over **Crazyflie CRTP** (10 Hz setpoints, watchdog cut-off). Led detection and flight software in a 4-member team; authored the **IEEE-format paper**.

Swach Mysuru Abhyan: Civic Detection Platform
2026 – Present

PyTorch · DeepLabV3 · YOLO11-seg · FastAPI · Flutter · OpenCV · Kaggle

- Built an **offline dashcam analytics platform**: DeepLabV3 waste segmentation with temporal stabilisation and an **explainable 0–100 cleanliness score**; evaluated on **1,456 frames** from an hour of dashcam video, with 0 false positives on 11 negative frames.
- Shipped a **FastAPI** service (async job queue, upload validation) and a **Flutter Windows / Android** client, backed by **26 automated test modules**; designed a 9-class **YOLO11-seg** pipeline and annotation kits for a 3-person team.

AI/ML Learning Platform · live on Vercel
Sep – Oct 2026

React · TypeScript · Tailwind · Supabase (PostgreSQL + Edge Functions) · TanStack Query

- Built a **gamified 30-day AI/ML course** (quizzes, in-browser Python challenges, streaks, rewards) with **8 serverless Edge Functions** verifying every score server-side, 9 SQL migrations and an admin activity dashboard.

Applied AI & Systems Builds
2026

- **ULTRON voice assistant:** wake word → STT → LLM → TTS with **speaker verification** and anti-hallucination guardrails.
- **Gesture control:** webcam-only mouse with MediaPipe Hands (drag, click, right-click) tuned on real hardware. **Quant V1:** PyTorch LSTM with testnet execution and 6-year backtests.

EDUCATION

B.Tech, Computer Science Engineering
Siddhartha Institute of Engineering & Technology, Hyderabad
Expected 2027 · CGPA 8.5 / 10

Diploma, Computer Science Engineering
Government Polytechnic Raichur, Karnataka
2021 – 2024 · Final CGPA 9.12 / 10

ACHIEVEMENTS & CERTIFICATIONS

- **1st Prize ×3:** INSPIRA '26 (IIT Hyderabad), SHINORA '26 (TKR College), AZURA '26 (CMR College)
- Flutter Development Certification, IVIS Labs (2023)
- IEEE-format research paper author · Languages: English, Telugu, Kannada, Hindi